Statistics: Introduction

- In the modern world of computers and information technology, the importance of statistics is very well recogonised by all the disciplines.
- Statistics has originated as a science of statehood and found applications slowly and steadily in Agriculture, Economics, Commerce, Biology, Medicine, Industry, planning, education and so on.

Definition of Statistics

- Statistics is defined differently by different authors over a period of time
- Statistics are numerical statement of facts in any department of enquiry placed in relation to each other. - A.L. Bowley
- Statistics may be defined as the science of collection, presentation analysis and interpretation of numerical data from the logical analysis. It is clear that the definition of statistics by Croxton and Cowden is the most scientific and realistic one. According to this definition there are four stages:Collection of Data,Presentation of data,Analysis of data and Interpretation of data. - Croxton and Cowden:

Data Analysis

- The data can be collected in connection with time or geographical location or in connection with time and location.
- Any statistical data can be classified under two categories depending upon the sources utilized.
- Primiary data
- Secondary data.

Primiary data

Primary data is the one, which is collected by the investigator himself for the purpose of a specific inquiry or study. Such data is original in character and is generated by survey conducted by individuals or research institution or any organisation

Example

If a researcher is interested to know the impact of noonmeal scheme for the school children, he has to undertake a survey and collect data on the opinion of parents and children by asking relevant questions. Such a data collected for the purpose is called primary data.

Methods for Collecting Primary data

The primary data can be collected by the following five methods.

1. Direct personal interviews
2. Indirect Oral interviews
3. Information from correspondents
4. Mailed questionnaire method
5. Schedules sent through enumerators.

Secondary data

Secondary data are those data which have been already collected and analysed by some earlier agency for its own use; and later the same data are used by a different agency.

Frequency distribution

Frequency distribution is a series when a number of observations with similar or closely related values are put in separate bunches or groups, each group being in order of magnitude in a series. It is simply a table in which the data are grouped into classes and the number of cases which fall in each class are recorded. It shows the frequency of occurrence of different values of a single Phenomenon.

Frequency distribution

A frequency distribution is constructed for three main reasons:

1. To facilitate the analysis of data.
2. To estimate frequencies of the unknown population distribution from the distribution of sample data and
3. To facilitate the computation of various statistical measures.

Raw data or Ungrouped data

The statistical data collected are generally raw data or ungrouped data.

Example

Let us consider the daily wages (in Rs) of 30 labourers in a factory. 800, 700, 550, 500, 600, 650, 400, 300, 800, 900, 750, 450, 350, 650, 700, 800, 820, 550, 650, 800, 600, 550, 380, 650, 750, 850, 900, 650, 450, 750.

Discrete (or) Ungrouped frequency distribution

In this form of distribution, the frequency refers to discrete value. Here the data are presented in a way that exact measurement of units are clearly indicated.

Example

In a survey of 40 families in a village, the number of children per family was recorded and the following data obtained.

1	0	3	2	1	5	6	2
2	1	0	3	4	2	1	6
3	2	1	5	3	3	2	4
2	2	3	0	2	1	4	5
3	3	4	4	1	2	4	5

Discrete frequency distribution.

Represent the data in the form of a discrete frequency distribution.

Number of Childern	Frequency
0	3
1	7
2	10
3	8
4	6
5	4
6	2
Total	40

Continuous frequency distribution

In this form of distribution refers to groups of values. This becomes necessary in the case of some variables which can take any fractional value and in which case an exact measurement is not possible. Hence a discrete variable can be presented in the form of a continuous frequency distribution.

Example

Wage distribution of 100 employees

Weekly wages (Rs)	Number of employees
50-100	4
100-150	12
150-200	22
200-250	33
250-300	16
300-350	8
Total	100

Measures of Central Tendency

1. Arithmetic Mean
2. Median
3. Mode

Arithmetic Mean or Average

- For ungrouped data,
$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$
- If $x_i | f_i, i = 1, 2, \ldots, n$ is the frequency distribution, then Arithmetic Mean (AM) or Average $\bar{x}$ is given by
$$\bar{x} = \frac{\sum_{i=1}^{n} f_i x_i}{N}$$
where $N = \sum_{i=1}^{n} f_i$. In case of continuous frequency distribution x_i is taken as the middle value of the corresponding interval.

Median

- Median of distribution is the value of the variable which divides it into two equal parts
- It is the value which exceeds and is exceeded by the same number of observations
- Median is the value such that the number of observations above it is equal to the number of observations below it
- The median is thus a positional average

- In case of ungrouped data, if the number of observations is odd then median is the middle value after the values have been arranged in ascending or descending order of magnitude.
- In case of even number of observations, there are two middle terms and median is obtained by taking the arithmetic mean of the middle terms.
- For example, the median of the value 25, 20, 15, 35, 18, i.e., 15, 18, 20, 25, 35 is 20

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- For example, the median of the value 25, 20, 15, 35, 18, i.e., 15, 18, 20, 25, 35 is 20
- The median of 8, 20, 50, 25, 15, 30, i.e., of 8, 15, 20, 25, 30, 50 is $\frac{1}{2}(20 + 25) = 22.5$.

Median of discrete frequency distribution

In case of discrete frequency distribution median is obtained by considering the cumulative frequencies. The steps for calculating median are given below:

1. Find $N/2$, where $N = \sum_i f_i$
2. See the (less than) cumulative frequency (cf.) just greater than $N/2$
3. The corresponding value of x is median

Example

Obtain the median for the following frequency distribution:

x:	1	2	3	4	5	6	7	8	9
f:	8	10	11	16	20	25	15	9	6

Solution:

x	f	c.f.
1	8	8
2	10	18
3	11	29
4	16	45
5	20	65
6	25	90
7	15	105
8	9	114
9	6	120

Hence, $N = 120 \implies N/2 = 60$. Cumulative frequency (cf.) just greater than $N/2$ is 65 and the value of x corresponding to 65 is 5. Therefore, median is 5.

Median of continuous frequency distribution

In the case of continuous frequency distribution, the class corresponding to the cf. just greater than $N/2$ is called the median class and the value of median is obtained by the following formula :

$$Median = l + \frac{h}{f}\left(\frac{N}{2} - c\right)$$

where l is the lower limit of the median class, f is the frequency of the median class, h is the magnitude of the median class, c is the cf. of the class preceding the median class, and $N = \sum_i f_i$

Example

Find the median wage of the following distribution:

Wages (in Rs.) :	20–30	30–40	40–50	50–60	60–70
No. of labourers :	3	5	20	10	5

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Solution:

Wages (in Rs.)	No. of labourers	c.f.
20–30	3	3
30–40	5	8
40–50	20	28
50–60	10	38
60–70	5	43

Here $N/2 = 43/2 = 21.5$.

Cumulative frequency just greater than 21.5 is 28 and the corresponding class is 40–50. Thus median class is 40–50.

$$Median = 40 + (10/20)(21.5 - 8) = 40 + 6.75 = 46.75.$$

Thus median wage is Rs. 46.75.

Example

An incomplete frequency distribution is given as follows

Variable	Frequency	Variable	Frequency
10–20	12	50-60	?
20–30	30	60 70	25
30–40	?	70-80	18
40–50	65	Total	229

Given that the median value is 46, determine the missing frequencies using the median formula.

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30–40	?	70-80	18
40–50	65	Total	229

Given that the median value is 46, determine the missing frequencies using the median formula.

Solution: Let the frequency of the class 30–40 be f_1 and that of 50–60 f_2. Then,

$$f_1 + f_2 = 229 - (12 + 30 + 65 + 25 + 18) = 79.$$

Since median is given to be 46, the class 40–50 is the median class. Hence using median formula we get

$$46 = 40 + \frac{114.5 - (12 + 30 + f_1)}{60} x10$$

which gives $f_1 = 34$ and $f_2 = 45$, since frequency never be fractional and $f_1 + f_2 = 79$.

Mode

- Mode is the value which occurs most frequently in a set of observations and around which the other items of the set cluster densely.
- Mode is the value of the variable which is predominant in the series.

Definition (Discrete frequency distribution)

In the case of discrete frequency distribution mode is the value of x corresponding to maximum frequency.

For example, in the following frequency distribution:

x:	1	2	3	4	5	6	7	8
f :	4	9	16	25	22	15	7	3

the value of x corresponding to the maximum frequency, viz., 25 is 4. Hence mode is 4.

But in anyone (or-more) of the following cases :

- if the maximum frequency is repeated
- if the maximum frequency occurs in the very beginning or at the end of the distribution and
- if there are irregularity in the distribution, the value of mode is determined by the method of grouping. which is illustrated below by an example.

Discrete frequency distribution

Example

Find the mode of the following frequency distribution:

Size (x):	1	2	3	4	5	6	7	8	9	10	11	12
Frequency (f):	3	8	15	23	35	40	32	28	20	45	14	6

Size (x)	Frequency (i)	(ii)	(iii)	(iv)	(v)	(vi)
1	3	11		26		
2	8		23		46	
3	15	38				73
4	23		58	98		
5	35	75			107	
6	40		72			100
7	32	60		80		
8	28		48		93	
9	20	65				79
10	45		59	65		
11	14	20				
12	6					

(a)

ANALYSIS TABLE

Column Number (1)	Maximum Frequency (2)	Value or combination of values of x giving max. frequency in (2) (3)
(i)	45	10
(ii)	75	5, 6
(iii)	72	6, 7
(iv)	98	4, 5, 6,
(v)	107	5, 6, 7
(vi)	100	6, 7, 8

(b)

Continuous frequency distribution

In case of continuous frequency distribution. Mode is given by the formula :

$$Mode = l + \frac{h(f_1 - f_0)}{2f_1 - f_0 - f_2}$$

Here l is the lower limit, h the magnitude and f_1 the frequency of the modal class, f_0 and f_2 are the frequencies of the classes preceding and succeeding the modal class respectively.

Example

Find the mode of the following frequency distribution:

Class-interval:	0–10	10–20	20–30	30–40	40–50	50–60	60–70
Frequency:	5	8	7	12	28	20	10

Example

Find the mode of the following frequency distribution:

Class-interval:	0–10	10–20	20–30	30–40	40–50	50–60	60–70
Frequency:	5	8	7	12	28	20	10

Solution: Maximum frequency is 28. Thus the class 40-50 is the modal class.

- $l = 40$, the lower limit of the modal class
- $h = 10$, the magnitude
- $f_1 = 28$, the frequency of the modal class
- $f_0 = 12$ and $f_2 = 20$

Answer$=$46.67 (approx.).

Example

The median and mode of the following wages distribution are known to be Rs.33.50 and Rs.34 respectively. Find the value of f_3, f_4, f_5.

Class-interval:	0–10	10–20	20–30	30–40	40–50	50–60	60–70
Frequency:	4	16	f_3	f_4	f_5	6	4

Solution:

CALCULATIONS FOR MODE AND MEDIAN

Wages (in Rs.)	*Frequency (f)*	*Less than c.f.*
0—10	4	4
10—20	16	20
20—30	f_3	$20+f_3$
30—40	f_4	$20+f_3+f_4$
40—50	f_5	$20+f_3+f_4+f_5$
50—60	6	$26+f_3+f_4+f_5$
60—70	4	$30+f_3+f_4+f_5$
Total	$230=30+f_3+f_4+f_5$	

Since median is 33.5, which lies in the class 30-40, 30-40 is the median class. Using the median formula $Median = l + \frac{h}{f}\left(\frac{N}{2} - c\right)$, we get

$$f_3 = 95 - 0.35 f_4$$

Mode being 34, the modal class is also 30-40. Using mode formula $Mode = l + \frac{h(f_4 - f_3)}{2f_4 - f_3 - f_5}$, we get

$$34 = 30 + \frac{10(f_4 - f_3)}{2f_4 - f_3 - f_5}.$$

By applying $f_3 = 95 - 0.35 f_4$ and $200 - f_4 = -f_3 - f_5$, we have $f_4 = 100$ $f_3 = 60, f_5 = 40$

Karl Pearson Relationship

Sometimes mode is estimated from the mean and the median. For a symmetrical distribution, mean, median and mode coincide. If the distribution is moderately asymmetrical, the mean, median and mode obey the following empirical relationship (due to Karl Pearson) :

The distance between mean and median is about one-third of the distance between the mean and mode

$$Mean - Median = \frac{1}{3}(Mean - Mode)$$

which gives $Mode = 3Median - 2Mean$

Relation between Mean, Median, Mode

1. In symmetrical distribution $Mean = Median = Mode$
2. In positively skewed distribution $Mode < Median < Mean$
3. In negatively skewed distribution $Mean < Median < Mode$

Geometric Mean and Harmonic Mean

Definition (Geometric Mean)

Geometric mean of a set of n observations is the n^{th} root of their product. Thus the geometric mean GM of n observations $x_i, i = 1, 2, \ldots, n$ is

$$GM = (x_1 x_2 \ldots x_n)^{1/n}$$

Definition (Harmonic Mean)

Harmonic mean of n number of observations is the reciprocal of the arithmetic mean of the reciprocals of the given values. Thus the harmonic mean HM of n observations $x_i, i = 1, 2, \ldots, n$ is

$$HM = \frac{1}{\frac{1}{n}\sum_{i=1}^{n}(1/x_i)}.$$

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Quartiles

- $$Q_1 = l + \frac{h}{f}\left(\frac{N}{4} - c\right)$$
- $$Q_2 = l + \frac{h}{f}\left(\frac{N}{2} - c\right)$$
- $$Q_3 = l + \frac{h}{f}\left(\frac{3N}{4} - c\right)$$

where l is the lower limit of the Q_i class, f is the frequency of the Q_i class, h is the magnitude of the Q_i class, c is the cf. of the class preceding the Q_i class, and $N = \sum_i f_i$

Measures of Dispersion

1. Range
2. Quartile deviation or Semi-interquartile range
3. Mean deviation
4. Standard deviation

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Definition (Quartile deviation)

Quartile deviation or semi-interquartile range Q is given by

$$Q = \frac{1}{2}(Q_3 - Q_1)$$

where Q_1 and Q_3 are the first and third quartiles of the distribution respectively.

Measures of Dispersion

Definition (Mean Deviation)

If $x_i | f_i$, $i = 1, 2, \dots, n$ is the frequency distribution, then mean deviation from the average $\hat{x}$, (usually mean, median or mode), is given by

$$Mean\ deviation = \frac{1}{N} \sum_{i}^{n} f_i |x_i - \bar{x}|$$

where $|x_i - \bar{x}|$ represents the modulus or the absolute value of the deviation $(x_i - A)$, when the -ve sign is ignored $\sum_i f_i = N$.

Measures of Dispersion

Definition (Standard Deviation)

If $x|f_i, i = 1, 2, \ldots, n$ is the frequency distribution, then standard deviation is given by

$$\sigma = \sqrt{\frac{1}{N}\sum_{i}^{n} f_i(x_i - \bar{x})^2}$$

where $\bar{x}$ represents the arithmetic mean of the distribution and $\sum_i f_i = N$.

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The square of standard deviation is called the variance and is given by

$$\sigma^2 = \frac{1}{N}\sum_{i}^{n} f_i(x_i - \bar{x})^2$$

Measures of Dispersion

Definition (Root mean square deviation)

If $x|f_i, i = 1, 2, \ldots, n$ is the frequency distribution, then the Root mean square deviation is given by

$$s = \sqrt{\frac{1}{N}\sum_{i}^{n} f_i(x_i - A)^2}$$

where A is any arbitrary number and $\sum_i f_i = N$.

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s^2 is called the mean square deviation.

Example

Calculate the Mean, Standard deviation for the following age distribution of 542 members

age in year:	20–30	30–40	40–50	50–60	60–70	70–80	80–90
#members:	3	61	132	153	140	51	2

Solution:

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Solution:

- Mean $\bar{x} = 54.72$ years
- Standard deviation $(\sigma) = 11.55$ years

Different formulae for calculating variance

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- $\sigma_x^2 = \frac{1}{N}\sum_i^n f_i d_i^2 - \left(\frac{1}{N}\sum_i^n f_i d_i\right)^2$, where $d_i = x_i - A, \bar{x} = A + \frac{1}{N}\sum_i^n f_i d_i$. In case of grouped- or continuous frequency distribution $d_i = \frac{x_i - A}{h}$, where h is common magnitude of class interval, A is arbitrary point.

Take $d = \frac{x-A}{h} = \frac{x-55}{10}$

Age group	Mid-value (x)	Frequency (f)	$d = \frac{x-55}{10}$	fd	fd^2
20–30	25	3	-3	-9	27
30–40	35	61	-2	-122	244
40–50	45	132	-1	-132	132
50–60	55	153	0	0	0
60–70	65	140	1	140	140
70–80	75	51	2	102	204
80–90	85	2	3	6	18
		542		-15	765

$$\bar{x} = A + h\frac{\sum fd}{N} = 55 + \frac{10\times -15}{542} = 54.72 \text{ years}$$

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$$\sigma_x^2 = h^2\left[\frac{1}{N}\sum_i^n f_i d_i^2 - \left(\frac{1}{N}\sum_i^n f_i d_i\right)^2\right] = 133.3$$

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Standard deviation (σ) =11.55 years

Co-efficient of Variation

100 times the co-efficient of dispersion based upon standard deviation is called co-efficient of variation (C. V.),

$$C.V = 100 \times \frac{\sigma}{\bar{x}}$$

Moments

The r^{th} moment of a variable x about any point $x = A$. usually denoted by μ'_r is given

$$\begin{aligned} \mu'_r &= \frac{1}{N}\sum_i f_i(x_i - A)^r \\ &= \frac{1}{N}\sum_i f_i d_i^r, \end{aligned}$$

where $d_i = x_i - A$. The r^{th} moment of a variable about the mean $\bar{x}$. usually denoted by μ_r is given by

$$\mu_r = \frac{1}{N}\sum_i f_i(x_i - \bar{x})^r$$

NOTE:

- $\mu_0 = 1$
- $\mu_1 = 0$
- $\mu_2 = \sigma^2$

Pearson's β and γ coefficients

Karl Pearson defined the following four coefficients, based upon the first four moments about mean:

- $\beta_1 = \frac{\mu_3^2}{\mu_2^3}$
- $\gamma_1 = +\sqrt{\beta_1}$
- $\beta_2 = \frac{\mu_4}{\mu_2^2}$
- $\gamma_2 = \beta_2 - 3$

Skewness β_1 and Kurtosis β_2

Skewness β_1

- In symmetrical distribution $\bar{x} = M_d = M_0$: $\beta_1 = 0$
- In positively skewed distribution $\bar{x} > M_d > M_0$: $\beta_1 > 0$
- In negatively skewed distribution $\bar{x} < M_d < M_0$: $\beta_1 < 0$

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Kurtosis β_2

- Mesokurtic: $\beta_2 = 3$
- Leptokurtic: $\beta_2 > 3$
- Playtkurtic; $\beta_2 < 3$

Summary

- $\bar{x} = \frac{\sum_{i=1}^{n} f_i x_i}{N}$,
- $Median = l + \frac{h}{f}\left(\frac{N}{2} - c\right)$
- $Mode = l + \frac{h(f_1 - f_0)}{2f_1 - f_0 - f_2}$
- $Q_i = l + \frac{h}{f}\left(\frac{i*N}{4} - c\right)$ and Quartile Deviation, $Q = \frac{1}{2}(Q_3 - Q_1)$
- Standard Deviation, $\sigma = \sqrt{\frac{1}{N}\sum_i^n f_i(x_i - \bar{x})^2}$
- $C.V = 100 \times \frac{\sigma}{\bar{x}}$

Summary

- $\sigma_x^2 = \frac{1}{N}\sum_i^n f_i d_i^2 - \left(\frac{1}{N}\sum_i^n f_i d_i\right)^2$, $d_i = x_i - A$.
- In continuous frequency distribution,
  $\sigma_x^2 = h^2\left[\frac{1}{N}\sum_i^n f_i d_i^2 - \left(\frac{1}{N}\sum_i^n f_i d_i\right)^2\right]$, where $d_i = \frac{x_i - A}{h}$
- $\bar{x} = A + \frac{1}{N}\sum_i^n f_i d_i$, $\bar{x} = A + \frac{h}{N}\sum_i^n f_i d_i$
- $C.V = 100 \times \frac{\sigma}{\bar{x}}$
- $s = \sqrt{\frac{1}{N}\sum_i^n f_i (x_i - A)^2}$
- $Mean\ deviation = \frac{1}{N}\sum_i^n f_i |x_i - \bar{x}|$

Problems

1. Find the mean, median, mode, variance, standard deviation, quartile deviation and co-efficient of variation (CV) for the following frequency distribution.

Wages (in Rs.):	170–180	180–190	190–200	200–210	210–220	220–230	230–240	240–250
No. of Persons:	52	68	85	92	100	95	70	28

2. Calculate the mean, variance and standard deviation for the following frequency distribution, and hence obtain the value of co-efficient of variation.

Size (x) :	0	1	2	3	4	5	6	7	8
Frequency (f):	1	8	28	56	70	56	28	8	1

3. Find the value of a, $P(X < 3)$, cumulative distribution function, mean, variance and standard deviation of the discrete random variable (X) with the following probability distribution.

$X = x$:	0	1	2	3	4	5	6	7	8
$p = P(X = x)$:	a	$3a$	$5a$	$7a$	$9a$	$11a$	$13a$	$15a$	$17a$

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